

SCENARIO: "Internet is not working"

If config:

```
vboxuser@Ubuntu:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe7f:18e9 prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:fe7f:18e9 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:7f:18:e9 txqueuelen 1000 (Ethernet)
    RX packets 16873 bytes 21673950 (21.6 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 11145 bytes 808069 (808.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 586 bytes 50300 (50.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 586 bytes 50300 (50.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

vboxuser@Ubuntu:~$
```

WHY WE USE IT

To check:

- IP address
- network adapter status
- whether system connected to network

Issue

Internet not working.

Command Used

ifconfig

Findings

Network interface was active and assigned IP address 10.0.2.15.

Conclusion

System successfully connected to network.

SCENARIO:

“Website is not opening”

PING :

```
vboxuser@Ubuntu:~$ ping google.com
PING google.com (142.251.222.110) 56(84) bytes of data.
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=1 ttl=64 time=255 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=2 ttl=64 time=281 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=3 ttl=64 time=275 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=4 ttl=64 time=271 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=5 ttl=64 time=266 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=6 ttl=64 time=273 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=7 ttl=64 time=111 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=8 ttl=64 time=263 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=9 ttl=64 time=262 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=10 ttl=64 time=268 ms
64 bytes from pnbomb-az-in-f14.1e100.net (142.251.222.110): icmp_seq=11 ttl=64 time=262 ms
```

WHY WE USE IT

Checks:

internet connectivity

DNS resolution

packet communication

Issue

Unable to access websites.

Command Used

ping google.com

Findings

DNS resolution failure detected.

Resolution

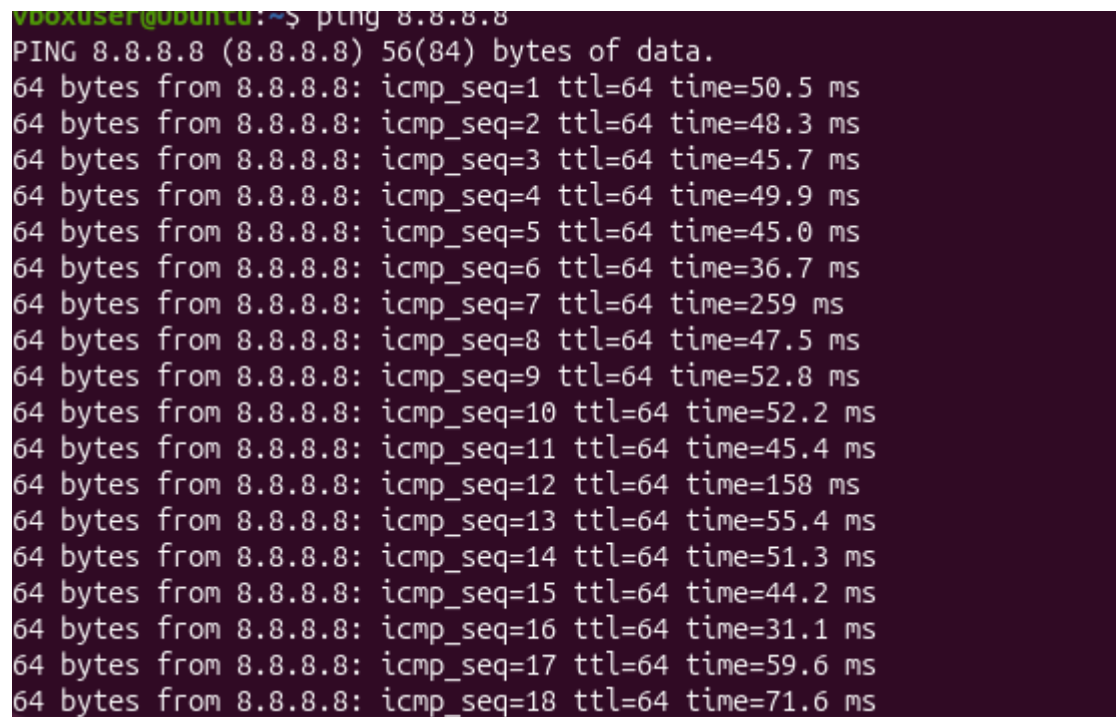
Checked DNS configuration.

Learning Outcome

Understood internet connectivity testing.

SCENARIO 3 — CHECK WHETHER INTERNET OR DNS ISSUE:

Ping 8.8.8.8:

A terminal window with a dark purple background. The prompt is 'vboxuser@ubuntu:~\$'. The command 'ping 8.8.8.8' has been entered. The output shows 'PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.' followed by 18 lines of ping results. Each line shows '64 bytes from 8.8.8.8: icmp_seq=X ttl=64 time=Y ms' where X ranges from 1 to 18 and Y represents the round-trip time in milliseconds.

```
vboxuser@ubuntu:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=50.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=48.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=45.7 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=49.9 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=64 time=45.0 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=64 time=36.7 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=64 time=259 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=64 time=47.5 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=64 time=52.8 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=64 time=52.2 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=64 time=45.4 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=64 time=158 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=64 time=55.4 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=64 time=51.3 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=64 time=44.2 ms
64 bytes from 8.8.8.8: icmp_seq=16 ttl=64 time=31.1 ms
64 bytes from 8.8.8.8: icmp_seq=17 ttl=64 time=59.6 ms
64 bytes from 8.8.8.8: icmp_seq=18 ttl=64 time=71.6 ms
```

Diagnostic Test

ping 8.8.8.8

Findings

IP connectivity successful.

Conclusion

Internet connectivity available but DNS resolution failed.

SCENARIO 4 — USER SAYS:

“Need to know system IP”

Hostname -I:

```
vboxuser@Ubuntu:~$ hostname -I
10.0.2.15 fd17:625c:f037:2:a00:27ff:fe7f:18e9
vboxuser@Ubuntu:~$
```

SCENARIO 5 — CHECK DNS INFORMATION

```
vboxuser@Ubuntu:~$ nslookup google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.221.238
Name:   google.com
Address: 2404:6800:4009:810::200e

vboxuser@Ubuntu:~$
```

WHY WE USE IT

Checks:

- DNS server
- domain resolution

OUTPUT MEANING

Address: 142.250.x.x

Means:

DNS converted:

google.com → IP address

Issue

Website hostname not resolving.

Command Used

nslookup google.com

Findings

DNS successfully resolved hostname to IP address.

Learning Outcome

Understood DNS resolution process.

SCENARIO 7 — CHECK OPEN CONNECTIONS:

```
boxuser@Ubuntu:~$ netstat -tulnp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp6       0      0 :::22                   :::*                     LISTEN
tcp6       0      0 :::1:631                 :::*                     LISTEN
udp        0      0 127.0.0.54:53           0.0.0.0:*
udp        0      0 127.0.0.53:53           0.0.0.0:*
udp        0      0 127.0.0.1:323           0.0.0.0:*
```

WHY WE USE IT

Shows:

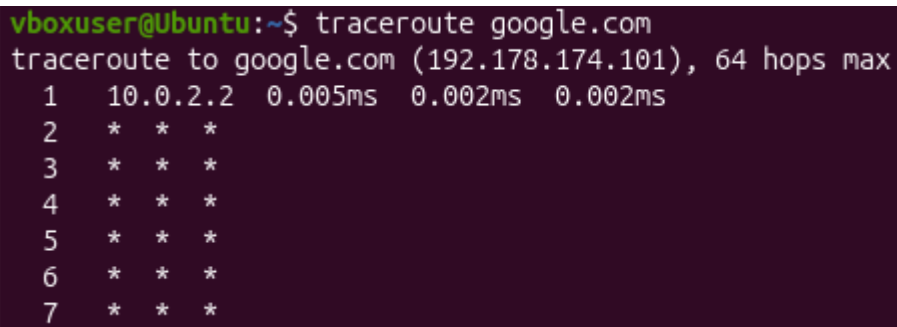
- listening ports
- active services
- network connections

SERVICE DESK USE

Useful when:

- application not responding
- server service issue
- port troubleshooting

SCENARIO 8 — CHECK NETWORK PATH:



```
vboxuser@Ubuntu:~$ traceroute google.com
traceroute to google.com (192.178.174.101), 64 hops max
 1  10.0.2.2  0.005ms  0.002ms  0.002ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
```

WHY WE USE IT

Shows:

route packets travel through network.

SERVICE DESK USE

Used when:

- internet slow
- routing problem
- packet loss issue

SCENARIO 9 — REMOTE LOGIN

ssh [user@192.168.1.10](#)

WHY WE USE IT

Allows:

remote system access.

SERVICE DESK USE

Very common in:

- Linux support
- server administration
- remote troubleshooting

SCENARIO 10 — COPY FILES REMOTELY

scp file.txt [user@192.168.1.10:/home/user](#)

WHY WE USE IT

Securely transfers files between systems.

SERVICE DESK USE

Used for:

- log transfer
- backup
- troubleshooting files

WINDOWS NETWORK TROUBLE SHOOTING COMMANDS :

hostname :Displays the computer name of the local machine. Identify the user's PC on the network.

Getmac: Displays the MAC (Physical) Address of network adapters. Used for device identification and network troubleshooting.

```
C:\Users\dexte>hostname
deXx

C:\Users\dexte>getmac

Physical Address      Transport Name
=====
10-68-38-8E-83-01    \Device\Tcpip_{227E783A-5750-400E-A8DE-85B96E1EFA0F}
08-BF-B8-68-70-01    Media disconnected
0A-00-27-00-00-12    \Device\Tcpip_{CDA3DFC9-73E4-4F0B-AFBE-7DF9F6A647CB}
```

**net
share**

Shows all shared folders and resources on the computer.

```
C:\Users\dexte>net share
```

Share name	Resource	Remark
C\$	C:\	Default share
IPC\$		Remote IPC
ADMIN\$	C:\WINDOWS	Remote Admin
Users	C:\Users	

The command completed successfully.

Ping :

Tests connectivity between your computer and another device/server.

ping google.com

ipconfig :

Displays IP address, subnet mask, and default gateway information.


```

C:\Users\dexte>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::a321:987e:98e4:28fe%18
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2409:40f4:3042:3718:e11b:9ea3:6939:53d6
    Temporary IPv6 Address. . . . . : 2409:40f4:3042:3718:2493:82ec:fb6c:4b2e
    Link-local IPv6 Address . . . . . : fe80::4804:61ff:c04f:56fe%6
    IPv4 Address. . . . . : 10.162.38.194
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::a835:ddff:fe08:a47d%6
                                10.162.38.226

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

```

Tracert :

Traces the route packets take to reach a destination.

Find where network delays occur.

```
Tracing route to google.com [2404:6800:4009:80b::2004]
over a maximum of 30 hops:
```

```
1      3 ms      2 ms      2 ms      ^C
C:\Users\dexte>tracert google.com
```

```
Tracing route to google.com [2404:6800:4000:1017::65]
over a maximum of 30 hops:
```

1	2 ms	2 ms	1 ms	2409:40f4:3159:97d8::a4
2	*	*	*	Request timed out.
3	60 ms	23 ms	25 ms	2405:200:5218:23:3925::ff24
4	*	*	*	Request timed out.
5	*	*	*	Request timed out.
6	35 ms	23 ms	23 ms	2405:200:801:4a00::ec
7	*	*	*	Request timed out.
8	*	*	*	Request timed out.
9	40 ms	43 ms	30 ms	2404:6800:8202:400::1
10	40 ms	33 ms	31 ms	2404:6800:8202:400::1
11	65 ms	32 ms	29 ms	2404:6800:8202:400::1
12	201 ms	35 ms	39 ms	2001:4860:0:1::1c74
13	32 ms	60 ms	32 ms	2001:4860:0:1::40e8

Ipconfig /all :

```
C:\Users\dexte>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : deXx
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-12
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a321:987e:98e4:28fe%18(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 738852903
DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-E9-A6-17-08-BF-B8-68-70-01
NetBIOS over Tcpip. . . . . : Enabled
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 12-68-38-8E-83-21
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 12-68-38-8E-83-31
DHCP Enabled. . . . . : Yes
```

Nslookup :

Checks DNS name resolution.

Verify whether DNS is resolving websites correctly.

ARP :

Displays the ARP cache (IP-to-MAC address mappings).

Troubleshoot LAN communication issues.

```
C:\Users\dexte>arp -a
```

```
Interface: 10.162.38.194 --- 0x6
```

Internet Address	Physical Address	Type
10.162.38.226	a6-27-e0-d9-a4-b8	dynamic
10.162.38.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
224.77.77.77	01-00-5e-4d-4d-4d	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
Interface: 192.168.56.1 --- 0x12
```

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
224.77.77.77	01-00-5e-4d-4d-4d	static
239.255.255.250	01-00-5e-7f-ff-fa	static

Netstat :

Displays active network connections and listening ports.

Check network sessions and ports.

Active Connections

Proto	Local Address	Foreign Address	State
TCP	10.162.38.194:49473	4.213.25.240:https	ESTABLISHED
TCP	10.162.38.194:49697	104.46.162.225:https	ESTABLISHED
TCP	10.162.38.194:64608	whatsapp-cdn-shv-01-del3:https	ESTABLISHED
TCP	127.0.0.1:1042	checkhost:56394	ESTABLISHED
TCP	127.0.0.1:1042	checkhost:56395	ESTABLISHED
TCP	127.0.0.1:6850	checkhost:58642	ESTABLISHED
TCP	127.0.0.1:9012	checkhost:56392	ESTABLISHED
TCP	127.0.0.1:13030	checkhost:49670	ESTABLISHED
TCP	127.0.0.1:13031	checkhost:49682	ESTABLISHED
TCP	127.0.0.1:49670	checkhost:13030	ESTABLISHED
TCP	127.0.0.1:49671	checkhost:49672	ESTABLISHED
TCP	127.0.0.1:49672	checkhost:49671	ESTABLISHED
TCP	127.0.0.1:49673	checkhost:49674	ESTABLISHED
TCP	127.0.0.1:49674	checkhost:49673	ESTABLISHED
TCP	127.0.0.1:49682	checkhost:13031	ESTABLISHED
TCP	127.0.0.1:50100	checkhost:56382	ESTABLISHED
TCP	127.0.0.1:56382	checkhost:50100	ESTABLISHED
TCP	127.0.0.1:56392	checkhost:9012	ESTABLISHED
TCP	127.0.0.1:56394	checkhost:1042	ESTABLISHED
TCP	127.0.0.1:56395	checkhost:1042	ESTABLISHED
TCP	127.0.0.1:58642	checkhost:6850	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:49315	[2603:1040:a06:6::2]:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:49474	[2603:1040:a06:6::2]:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:49944	207:https	CLOSE_WAIT
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:50206	lb-140-82-114-22-iad:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:51052	[2606:4700:9c62:7dba:1add:aa0:5ff2:75c4]:htt	ESTABLISHED
ps	ESTABLISHED		
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:52433	se-in-f119:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:54212	sg-in-f188:5228	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:54214	[2603:1063:2001:1d2f::365:ff1]:https	ESTABLISHED
ISHED			
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:55184	se-in-f119:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:55185	[64:ff9b::142a:415e]:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:55966	[64:ff9b::3468:4535]:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:55967	[64:ff9b::1434:40c9]:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:60806	lb-140-82-112-26-iad:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:62766	[2602:f79a::2]:https	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:64043	sg-in-f188:5228	ESTABLISHED
TCP	[2409:40f4:3159:97d8:119d:9deb:9e5b:5332]:65071	[2600:1901:0:5e8a::]:https	CLOSE_WAIT

Tasklist :

C:\Users\dexte>tasklist

Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	1,396 K
Secure System	188	Services	0	43,384 K
Registry	232	Services	0	19,176 K
smss.exe	772	Services	0	N/A
csrss.exe	1164	Services	0	1,184 K
wininit.exe	1272	Services	0	476 K
services.exe	1348	Services	0	6,748 K
LsaIso.exe	1392	Services	0	4 K
lsass.exe	1400	Services	0	13,360 K
svchost.exe	1532	Services	0	27,972 K
fontdrvhost.exe	1560	Services	0	N/A
WUDFHost.exe	1616	Services	0	4,216 K
svchost.exe	1668	Services	0	17,908 K
WUDFHost.exe	1716	Services	0	4 K
svchost.exe	1852	Services	0	2,788 K
svchost.exe	2052	Services	0	880 K
svchost.exe	2068	Services	0	2,152 K

Taskkill :

Will kill the task with mentioning the pid

System info :

Displays detailed system information.

```
C:\Users\dexte>systeminfo

Host Name:                             DEXX
OS Name:                               Microsoft Windows 11 Home Single Language
OS Version:                             10.0.26200 N/A Build 26200
OS Manufacturer:                       Microsoft Corporation
OS Configuration:                      Standalone Workstation
OS Build Type:                           Multiprocessor Free
Registered Owner:                       dextertt17@gmail.com
Registered Organization:                 N/A
Product ID:                             00356-24666-31533-AAOEM
Original Install Date:                   20-10-2024, 01:36:36
System Boot Time:                        02-06-2026, 10:44:24
System Manufacturer:                    ASUSTeK COMPUTER INC.
System Model:                            ASUS TUF Gaming F17 FX706HF_FX706HF
System Type:                             x64-based PC
Processor(s):                            1 Processor(s) Installed.
                                           [01]: Intel64 Family 6 Model 141 Stepping 1 GenuineIntel ~2688 Mhz
BIOS Version:                           American Megatrends International, LLC. FX706HF.312, 05-07-2024
Windows Directory:                      C:\WINDOWS
System Directory:                        C:\WINDOWS\system32
Boot Device:                             \Device\HarddiskVolume1
System Locale:                            en-us;English (United States)
Input Locale:                             00004009
Time Zone:                               (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:                   7,915 MB
Available Physical Memory:                949 MB
Virtual Memory: Max Size:                 17,236 MB
Virtual Memory: Available:                 1,894 MB
Virtual Memory: In Use:                   15,342 MB
Page File Location(s):                   C:\pagefile.sys
Domain:                                  WORKGROUP
Logon Server:                            \\DEXX
Hotfix(s):                               4 Hotfix(s) Installed.
                                           [01]: KB5092427
                                           [02]: KB5054156
                                           [03]: KB5089573
                                           [04]: KB5092734
Network Card(s):                         3 NIC(s) Installed.
                                           [01]: MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
                                           Connection Name: Wi-Fi
                                           DHCP Enabled:    Yes
```

Msconfig :

Opens System Configuration Utility.

Manage startup programs and boot options

```
C:\Users\dexte>net user
```

```
User accounts for \\DEXX
```

```
-----
Administrator          CodexSandboxOffline   CodexSandboxOnline
DefaultAccount          dexte                 Guest
WDAGUtilityAccount      WsiAccount
The command completed successfully.
```

Compmgmt.msc: Opens Computer Management console.

Route print :

```
=====
Interface List
```

```
18...0a 00 27 00 00 12 .....VirtualBox Host-Only Ethernet Adapter
 4...12 68 38 8e 83 21 .....Microsoft Wi-Fi Direct Virtual Adapter
 3...12 68 38 8e 83 31 .....Microsoft Wi-Fi Direct Virtual Adapter #2
 6...10 68 38 8e 83 01 .....MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
 9...08 bf b8 68 70 01 .....Realtek PCIe GbE Family Controller
 1.....Software Loopback Interface 1
=====
```

```
IPv4 Route Table
=====
```

```
Active Routes:
```

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	10.162.38.226	10.162.38.194	45
10.162.38.0	255.255.255.0	On-link	10.162.38.194	301
10.162.38.194	255.255.255.255	On-link	10.162.38.194	301
10.162.38.255	255.255.255.255	On-link	10.162.38.194	301
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331
192.168.56.0	255.255.255.0	On-link	192.168.56.1	281
192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	10.162.38.194	301
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
255.255.255.255	255.255.255.255	On-link	10.162.38.194	301

```
=====
Persistent Routes:
```

```
None
```

```
IPv6 Route Table
=====
```

```
Active Routes:
```

If	Metric	Network Destination	Gateway
6	61	::/0	fe80::a427:e0ff:fed9:a4b8
1	331	::1/128	On-link
6	61	2409:40f4:3159:97d8::/64	On-link
6	301	2409:40f4:3159:97d8:62:23e3:cfd:5357/128	On-link

We can add user with command called net user jameel/add

We can also delete user with this command Jameel/del

Traceroute :

```
C:\Users\dexte>route print
=====
Interface List
18...0a 00 27 00 00 12 .....VirtualBox Host-Only Ethernet Adapter
4...12 68 38 8e 83 21 .....Microsoft Wi-Fi Direct Virtual Adapter
3...12 68 38 8e 83 31 .....Microsoft Wi-Fi Direct Virtual Adapter #2
6...10 68 38 8e 83 01 .....MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
9...08 bf b8 68 70 01 .....Realtek PCIe GbE Family Controller
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          10.162.38.226    10.162.38.194    45
10.162.38.0                255.255.255.0    On-link          10.162.38.194    301
10.162.38.194              255.255.255.255  On-link          10.162.38.194    301
10.162.38.255              255.255.255.255  On-link          10.162.38.194    301
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
192.168.56.0                255.255.255.0    On-link          192.168.56.1     281
192.168.56.1                255.255.255.255  On-link          192.168.56.1     281
192.168.56.255             255.255.255.255  On-link          192.168.56.1     281
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link          192.168.56.1     281
224.0.0.0                  240.0.0.0        On-link          10.162.38.194    301
255.255.255.255            255.255.255.255  On-link          127.0.0.1        331
255.255.255.255            255.255.255.255  On-link          192.168.56.1     281
255.255.255.255            255.255.255.255  On-link          10.162.38.194    301
=====
```

Pathping tracert route : Combines ping and tracert to identify packet loss across network hops.

```
C:\Users\dexte>pathping google.com

Tracing route to google.com [2404:6800:4009:824::200e]
over a maximum of 30 hops:
 0 deXx [2409:40f4:3159:97d8:119d:9deb:9e5b:5332]
 1 2409:40f4:3159:97d8::a4
 2 * * *
Computing statistics for 25 seconds...
Source to Here This Node/Link
Hop RTT Lost/Sent = Pct Lost/Sent = Pct Address
0 deXx [2409:40f4:3159:97d8:119d:9deb:9e5b:5332]
1 79ms 0/ 100 = 0% 0/ 100 = 0% 2409:40f4:3159:97d8::a4
Trace complete.
```

Mste :

Opens Remote Desktop Connection (RDP).

Firewall .cpl

Opens Windows Defender Firewall settings.

: